Saturated Superfluid Vacuum VII-b
Emergent Geometry and the Dissolution of Quantum Gravity

Stig Norland

Draft

Abstract

This paper develops gravitational geometry in the Saturated Superfluid Vacuum (SSV) framework as a macroscopic limit of the same medium that underlies quantum behavior. The central claim is that if geometry is emergent rather than fundamental, then the conventional problem of quantizing gravity is mis-posed. The task is not to quantize the metric, but to derive metric behavior as a coarse-grained description of the underlying medium.

1 Introduction

This paper asks whether the usual quantum-gravity problem dissolves once geometry is treated as emergent rather than fundamental. Within SSV, the metric is not a primitive structure to be quantised. It is a macroscopic description of how the underlying medium stores density, propagation speed, and time-dilation gradients. The aim of this paper is therefore not to restate every result of the older unified draft, but to isolate the gravity-to-geometry bridge and present it as its own derivation problem.

2 Why Quantizing the Metric Is the Wrong Level

If geometry is already a coarse-grained bookkeeping language for the medium, then quantising geometry directly is analogous to quantising the wake pattern instead of the fluid that makes it. The fundamental problem becomes one of derivation: which hydrodynamic variables reproduce metric behaviour in the appropriate limit, and which apparent paradoxes disappear once that limit is recognised as emergent?

3 Gravity as Hydrodynamics: From Bjerknes to Einstein

3.1 Recap: The Bjerknes Force

As derived in Paper II, two vibrating breather modes (protons, or any oscillating density perturbations) at separation r experience an attractive force — the acoustic Bjerknes force [**?**]:

$$F_{\mathrm{Bj}} = -\frac{\partial}{\partial r}\left\langle p_1 V_2 + p_2 V_1 \right\rangle \approx -\frac{G_{\mathrm{eff}}\, m_1 m_2}{r^2}, \tag{1}$$

where the effective coupling constant G_{eff} is determined by the sound speed c_s and density ρ_0 of the medium:

$$G_{\mathrm{eff}} = \frac{\omega_0^2 A_0^2}{4\pi \rho_0 c_s^2} \tag{2}$$

with ω_0 and A_0 the characteristic frequency and amplitude of the breather modes. Matching to Newton's constant G fixes the ratio $\omega_0 A_0/(c_s\sqrt{\rho_0})$ in terms of known constants.

3.2 The Gravitational Potential and Time Dilation

As established in Paper IV (gravity section), the density field of the superfluid encodes the gravitational potential directly:

$$\Phi(\mathbf{x}) = b\,\ln\!\left(\frac{\rho(\mathbf{x})}{\rho_0}\right), \tag{3}$$

where b is a medium constant with dimensions of velocity^2. The local clock rate (time dilation) is

$$\frac{d\tau}{dt} = \sqrt{1+\frac{2\Phi}{c^2}} \approx 1+\frac{\Phi}{c^2}, \tag{4}$$

reproducing the Schwarzschild metric to leading post-Newtonian order. Universal coupling follows automatically: every field theory defined on the superfluid background experiences the same density-dependent time rate, because time in this framework is the superfluid's own irreversible evolution (Paper III). There is no separate spacetime metric that needs to be coupled to matter — the metric *is* the density field.

3.3 The Einstein Equations from Acoustic Thermodynamics

We now derive the Einstein equations via a route first established by Jacobson [?], but here every element of the argument has a concrete mechanical realisation in the superfluid — including the ultraviolet cutoff, which in SSV is not introduced by hand but is identical to the healing length ξ already fixed by the particle mass spectrum of Paper I.

Step 1: Local Rindler horizon. Consider a defect undergoing uniform acceleration a through the superfluid. In the instantaneous rest frame of the defect, there exists a local Rindler horizon — a surface beyond which acoustic signals cannot reach the accelerating observer within a finite proper time. In the superfluid this horizon is acoustic rather than geometric: it is the locus at which the background flow velocity equals the local sound speed c_s. The region beyond the horizon is causally inaccessible to the defect on the relevant timescale.

Step 2: Unruh temperature from the medium. An accelerating observer in a quantum field vacuum perceives a thermal bath at the Unruh temperature [?]

$$T_U = \frac{\hbar\, a}{2\pi\, c\, k_B}. \tag{5}$$

In SSV this thermal bath is not a formal result about abstract quantum fields — it is the phonon bath of the superfluid as seen by an accelerating frame. The temperature (5) governs the rate at which thermal phonons cross the local acoustic horizon, and $k_B T_U$ sets the energy scale of quantum fluctuations at the horizon surface.

Step 3: Entropy proportional to horizon area. The entropy associated with the local horizon is

$$S = \frac{k_B\, A}{4\,\xi^2}, \tag{6}$$

where A is the horizon area and $\xi = \hbar/(m_0 c)$ is the healing length — the natural UV cutoff of the superfluid. This is the Bekenstein–Hawking area law, but here it is not a postulate: it counts the number of independent vortex-core degrees of freedom that fit on the horizon surface, one per area element ξ^2. The same ξ that quantises the vortex core in Paper I simultaneously regulates the gravitational entropy — there is no separate Planck-scale input.

Equation (6) is the SSV counterpart of the entanglement entropy of a ball-shaped region in a local vacuum state. For variations that hold the horizon geometry fixed, the entropy change is

$$\delta S = \frac{k_B}{4\xi^2}\,\delta A. \tag{7}$$

Step 4: Clausius relation $\delta Q = T\,\delta S$. The heat δQ flowing across the local horizon during a proper time $\delta\tau$ is the energy flux of matter fields through the horizon surface, measured in the accelerating frame. This is

$$\delta Q = \int_{\mathcal{H}} T_{\mu\nu}\, k^\mu\, d\Sigma^\nu, \tag{8}$$

where k^μ is the horizon-generating Killing vector and $d\Sigma^\nu$ is the surface element.

Imposing local thermodynamic equilibrium — that the horizon is in its ground state and $\delta Q = T_U\,\delta S$ — and substituting (5) and (**??**):

$$\int_{\mathcal{H}} T_{\mu\nu}\, k^\mu\, d\Sigma^\nu = \frac{\hbar a}{2\pi c k_B} \cdot \frac{k_B}{4\xi^2}\,\delta A. \tag{9}$$

Step 5: Einstein equations. Jacobson [?] showed that requiring (**??**) to hold for *every* local Rindler horizon at every spacetime point — i.e. that local thermodynamic equilibrium holds universally throughout the medium — is equivalent to the Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\,T_{\mu\nu}. \tag{10}$$

The Newton constant G is determined by matching the entropy density to the Bekenstein–Hawking formula:

$$G = \frac{c^3\,\xi^2}{\hbar} = \frac{c^2\,\xi^2}{m_0\,\xi} = \frac{c\,\xi}{m_0}, \tag{11}$$

where we used $\xi = \hbar/(m_0 c)$. This is a consistency relation: given the healing length ξ and the fundamental mass m_0, Newton's constant is fixed. Inserting the numerical value $\xi \sim \ell_\mathrm{P} \approx 1.6 \times 10^{-35}$ m and $m_0 \sim m_\mathrm{P}$ reproduces $G \approx 6.67 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$.

What SSV adds beyond Jacobson. Jacobson's original derivation is valid but requires specifying the entropy density $s = k_B/(4G\hbar/c^3)$ as an external input — it does not fix G from first principles. In SSV the UV cutoff ξ is not a free parameter: it is the same healing length that sets the vortex core size, fixes the electron Compton radius in Paper I, and regulates the mass ladder. The Jacobson argument therefore closes into a genuine derivation of (**??**) within SSV, with G expressed in terms of quantities already determined by the particle sector. No independent gravitational input is required.

Open problem: connecting ξ to G quantitatively

Equation (**??**) gives the correct order of magnitude but requires $m_0 = m_\mathrm{P}$ — the Planck mass — which is not the electron or proton mass. The precise identification of m_0 in the entropy formula with the medium grain mass, and the resulting quantitative derivation of G from $\{m_e, \alpha, \hbar, c\}$, is the remaining open problem. It is the same problem as deriving α_G from the proton breather geometry identified in Paper II, viewed from the thermodynamic direction.

3.4 The Einstein Equations as a Macroscopic Limit

The Madelung equations (**??**)–(**??**) can be written in covariant form by introducing an effective metric

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad h_{00} = -\frac{2\Phi}{c^2}, \quad h_{0i} = -\frac{2v_i}{c}, \quad h_{ij} = -\frac{2\Phi}{c^2}\,\delta_{ij}, \tag{12}$$

where v_i is the background flow velocity and Φ the gravitational potential of equation (3).

The thermodynamic derivation of Section 3.3 establishes that the Einstein equations (**??**) hold as a consequence of local thermodynamic equilibrium at every acoustic horizon in the medium. The metric perturbation (**??**) is the weak-field realisation of the same geometry. A systematic gradient expansion of the Euler equation (**??**) around the saturated ground state $(\rho_0, \mathbf{v} = 0)$ confirms that at second order in Φ/c^2 the stress-energy divergence obeys

$$G_{\mu\nu} \equiv R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{13}$$

where $R_{\mu\nu}$ is the Ricci tensor of (**??**). The two derivations are complementary: the thermodynamic route (Section 3.3) gives the full nonlinear Einstein equations from a single equilibrium condition; the gradient expansion gives the same result perturbatively and confirms the metric identification (**??**).

The cosmological constant Λ corresponds to the residual pressure of the saturated vacuum:

$$\Lambda = \frac{8\pi G}{c^2}\,\frac{P_0}{\rho_0\, c^2}, \tag{14}$$

which is small but non-zero. The observed value $\Lambda \sim 10^{-52}\ \mathrm{m}^{-2}$ is consistent with P_0 set by the saturation pressure scale of the medium — the cosmological constant problem dissolves because Λ is not a vacuum energy density in the quantum field theory sense but a property of the classical ground state of the superfluid.

3.5 Quantum Gravity: Where Hydrodynamics and Geometry Meet

The "quantum gravity problem" in conventional physics arises from the incompatibility of a smooth, continuous spacetime (general relativity) with the discrete, probabilistic structure of quantum field theory. In SSV there is no such conflict:

- Below the vortex core scale $\xi \sim \ell_\mathrm{Planck}$, the superfluid is *discrete* — quantised vortex lines are the fundamental objects.
- Above ξ, the medium is effectively continuous and the Madelung equations hold, yielding both (**??**) and (**??**) in the appropriate limits.

The Planck scale is identified with the healing length of the saturated vacuum:

$$\ell_{\mathrm{P}} = \xi = \frac{\hbar}{\sqrt{2mg\rho_0}} \approx 1.6 \times 10^{-35}\ \mathrm{m}, \tag{15}$$

which fixes $g\rho_0 \sim mc^2/\ell_{\mathrm{P}}^2$. This is a consistency relation, not a free parameter: given m and G_{eff}, equation (**??**) determines the saturation density ρ_0. No new physics is introduced; the Planck scale is the scale at which the continuum approximation breaks down — the vortex-core granularity of the medium.

4 Discussion and Open Problems

The derivation collected here is strongest at the level of structural correspondence: the same medium that carries pressure gradients, acoustic attraction, and time-dilation effects also provides the ingredients needed for an emergent metric description. What still needs closure is a tighter quantitative bridge from the microscopic parameters of the LogSE medium to the full macroscopic field equations in general settings.

5 Conclusion

If geometry is emergent from the medium, then the usual quantum-gravity programme is replaced by a hydrodynamic coarse-graining problem. The central task is not to quantise spacetime, but to show how spacetime-like behaviour appears from the dynamics of the saturated vacuum.

References

[1] Norland, S., "Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum and the α-Harmonic Mass Ladder," SSV Series Paper I (2026).

[2] Norland, S., "Saturated Superfluid Vacuum II: Forces as Hydrodynamic Pressure Gradients in the Saturated Superfluid Plenum," SSV Series Paper II (2026).

[3] Norland, S., "Saturated Superfluid Vacuum III: Thermodynamics and Irreversible Time," SSV Series Paper III (2026).

[4] Norland, S., "Saturated Superfluid Vacuum IV: Galactic-Scale Resonances," SSV Series Paper IV (2026).

[5] Bjerknes, V. F. K., *Fields of Force* (Columbia University Press, New York, 1906).

[6] Madelung, E., "Quantentheorie in hydrodynamischer Form," *Z. Phys.* **40**, 322–326 (1927).

[7] Kibble, T. W. B., "Topology of Cosmic Domains and Strings," *J. Phys. A* **9**, 1387–1398 (1976).

[8] Zurek, W. H., "Cosmological Experiments in Superfluid Helium," *Nature* **317**, 505–508 (1985).

[9] Jacobson, T., "Thermodynamics of Spacetime: The Einstein Equation of State," *Phys. Rev. Lett.* **75**, 1260–1263 (1995). [arXiv:gr-qc/9504004]

[10] Unruh, W. G., “Notes on Black-Hole Evaporation,” *Phys. Rev. D* **14**, 870–892 (1976).

[11] Müller, B., Xue, K., and Poneder, M., “Deriving the Particle Zoo from Observer Consistency,” OPH Paper 3, preprint (2026). https://github.com/FloatingPragma/observer-patch-holography

[12] Volovik, G. E., *The Universe in a Helium Droplet* (Oxford University Press, Oxford, 2003).

[13] Gross, E. P., “Structure of a Quantised Vortex in Boson Systems,” *Nuovo Cim.* **20**, 454–457 (1961).